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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (Machine Design) (CBCS) Degree Examinations

May – June 2024

22MMD212 – EXPERIMENTAL STRESS ANALYSIS

Duration: 3 Hours

Max. Marks:100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

❖ Missing data, if any may be suitably assumed.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Derive the equations of equilibrium for the state of stress at a point.	10	L*1	1	1
b)	For the state of stress given below find the normal shearing and resultant stresses on the plane whose outer normal has direction cosines $\cos(n,x)=0.25$, $\cos(n,y)=0.5$. The state of stress is given by $\sigma_{xx}=200\text{MPa}$, $\sigma_{yy}=100\text{MPa}$, $\sigma_{zz}=250\text{MPa}$, $\tau_{xy}=175\text{MPa}$, $\tau_{yz}=100\text{MPa}$, $\tau_{zx}=225\text{MPa}$.	10	L3	1	1
2. a)	Derive an expression for Principal Strains and stresses in a three-element delta rosette.	10	L2	2	1
b)	At a point in a stressed body, the cartesian components of stress are $\sigma_{xx}=-800\text{KPa}$, $\sigma_{yy}=1200\text{KPa}$, $\sigma_{zz}=-400\text{KPa}$, $\tau_{xy}=400\text{KPa}$, $\tau_{yz}=-600\text{KPa}$, $\tau_{zx}=500\text{KPa}$. Determine a) Resultant stress, Normal and shear stress on a plane whose outer normal has direction cosines $\cos(n,x)=0.25$, $\cos(n,y)=0.5$.	10	L3	2	1
3. a)	Explain with neat sketches of wire type and foil type strain gauges and advantages of foil gauges over wire gauges.	10	L2	2	1
b)	The following readings of strain were obtained on a rectangular rosette mounted on Aluminium for which $E=70\text{GPa}$, Poissons ratio $\nu=0.32$. $\epsilon_a=285\mu$, $\epsilon_b=65\mu$, $\epsilon_c=102\mu$. Determine i) Principal Strains ii) Principal Stresses	10	L3	2	1
Unit – II					
4. a)	Explain i) Isochromatics ii) Isoclinics iii) Plane polarized light iv) Circularly polarized light v) Wave theory of light	5	L2	3	1
b)	Illustrate the application of Tardy's method of compensation in 2D photoelasticity to determine fractional fringe order.	5	L3	3	2
c)	Derive stress optic law for 2 dimensional photoelasticity.	10	L2	3	1
5. a)	How do you calibrate brittle coatings?	6	L1	4	1
b)	Discuss about crack detection methods in brittle coating	8	L2	4	1
c)	Discuss about ceramic based coatings?	6	L2	4	1
6. a)	Derive an expression for intensity of the light coming out of the analyser when a stressed photoelastic model is kept in a plane polariscope set for dark field.	10	L3	3	1
b)	How do you bond photo elastic coatings?	4	L2	4	1
c)	Write a note on Bire fringent coatings. What are the advantages of bire fringent method over other methods?	6	L2	4	1
Unit – III					
7. a)	Explain the geometric approach for Moire Fringe Analysis.	10	L2	5	1
b)	Explain the displacement approach for Moire Fringe Analysis.	10	L2	5	2
8. a)	Differentiate between spatial and temporal coherence.	10	L2	5	1
b)	What are the requirements for making a good quality hologram.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
